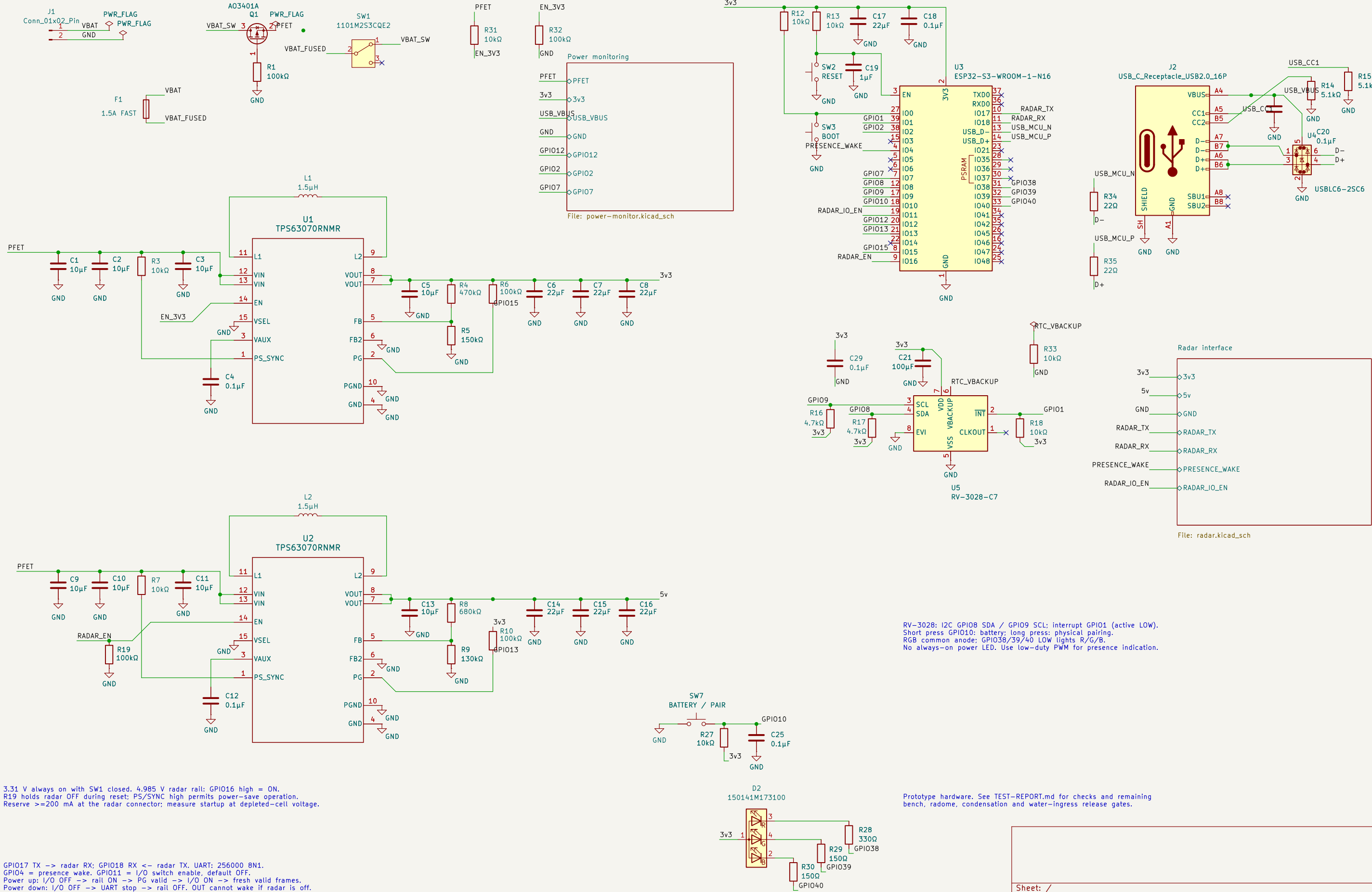


ALEC SENSOR S1 – prototype; bench and ingress tests required.
3 AA primary cells; no charging. USB-C is data-only; batteries + SW1 ON required.
SW1: hard battery disconnect, 6 A / 28 VDC. F1: 1.5 A fast fuse: validate inrush and fault clearing.
RTC time survives ESP deep sleep, but is lost on SW1 OFF or battery removal. Resync before arming.



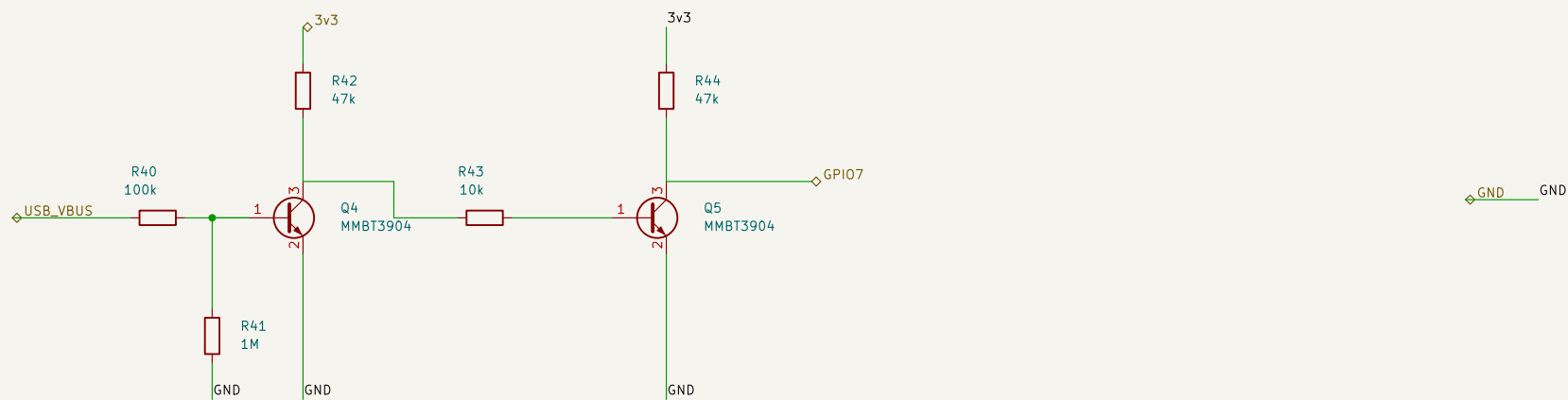
3.31 V always on with SW1 closed. 4.985 V radar rail: GPIO16 high = ON.
R19 holds radar OFF during reset; PS/SYNC high permits power-save operation.
Reserve >=200 mA at the radar connector; measure startup at depleted-cell voltage.

GPIO17 TX -> radar RX; GPIO18 RX <- radar TX. UART: 256000 8N1.
GPIO4 = presence wake. GPIO11 = I/O valid -> fresh valid frames.
Power up: I/O OFF -> rail ON -> I/O ON -> fresh valid frames.
Power down: I/O OFF -> UART stop -> rail OFF. OUT cannot wake if radar is off.

Sheet: /		
File: alec-sensor.kicad_sch		
Title: ALEC Sensor – prototype S1		
Size: A3	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 1/3

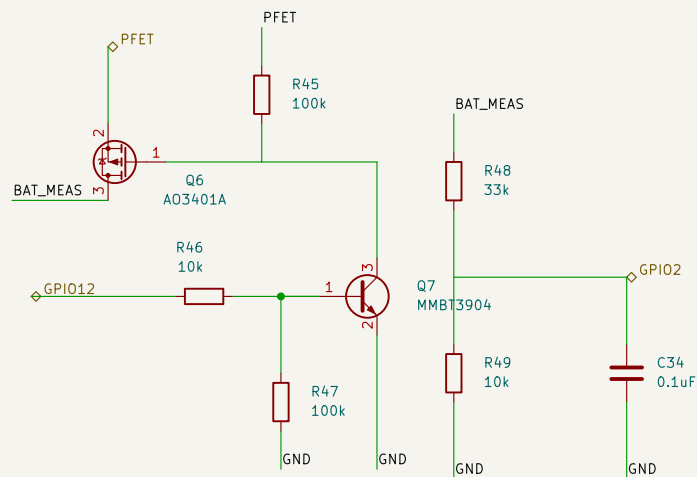
USB VBUS presence – HIGH only with a host supply connected

Two NPN stages keep VBUS away from the GPIO and preserve positive logic.
GPIO7: TinyUSB self_powered=true, vbus_monitor_io=7.
This is cable presence detection, not a precision USB undervoltage comparator.



Switched battery measurement – protected battery node to ADC1 / GPIO2

GPIO12 HIGH enables measurement; wait 10 ms, read calibrated ADC, then drive LOW.
Scale = 4.3; 33k / 10k, 0.1%; 100 nF filter, time constant 0.77 ms.
Designed measurement range 0-6 V; battery chemistry and percentage thresholds remain firmware settings.



Sheet: /Power monitoring/ File: power-monitor.kicad_sch		
Title: ALEC Sensor – power monitoring		
Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 2/3

